

Hallucination Removal via Chain Complex Homology

One-Page Technical Summary

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Core claim. Hallucination is not a single event. It is the intersection of failures in *global consistency*, *structural identifiability*, and *generative stability* — three mathematically distinct failure modes of a structured semantic transition system, each detectable by a different algebraic invariant.

The three failure modes

	Failure mode	Algebraic detector	Corresponds to
(i)	Global inconsistency	$H_1(C_*; \mathbb{Z}_5)$: cycle does not bound a filled 2-cell; $[\gamma] \neq 0$	Logical contradiction; impossible world state; "sounds right locally but cannot complete globally"
(ii)	Structural ambiguity	$\mathbb{Z}_2$ projection: orientation collapse; reversed path $\equiv$ forward path	Symmetry confusion; conflated meanings; indistinguishable reasoning paths
(iii)	Generative instability	2-adic v_2 (prefix): boundary mismatch accumulates in partial path	Semantic drift; compounding error; early divergence from admissible structure

Operational definition

$$\text{hallucination} \iff \underbrace{\neg(\text{stable prefix})}_{\text{(iii) 2-adic}} \vee \underbrace{\neg(\text{structurally distinct path})}_{\text{(ii) } \mathbb{Z}_2} \vee \underbrace{\neg(\text{globally fillable cycle})}_{\text{(i) } H_1(\mathbb{Z}_5)}$$

What the algebra actually does (important correction from experiment)

	Role	NOT
$H_1(C_*; \mathbb{Z}_5)$	Global admissibility classifier. $[\gamma] = 0$ iff cycle bounds a filled disk.	Not a "detector of truth." Classifies trajectory structure, not facts.
$\mathbb{Z}_2$	Orientation-forgetting projection of the same topology. $-1 \equiv +1$: direction disappears.	Not a weaker version of $\mathbb{Z}_5$. Different invariant, different question.
2-adic v_2	Prefix stability metric during generation. Meaningful only on partial paths, not completed cycles.	Not a topological invariant. Measures dynamics, not structure.

Empirical evidence

- **6-node chain complex.** $d^2 = 0$ verified over $\mathbb{Z}_2$ and $\mathbb{Z}_5$. Cycle A (filled 2-cell): $[\gamma_A] = 0$, gate allows. Cycle B (hole): $[\gamma_B] \neq 0$, gate blocks. Anti-FSM result: cycles share edges e_{23}, e_{34} — same adjacency, different homology. FSM cannot distinguish them; H_1 can.
- **Protocol benchmark.** Baseline nanoGPT: 54% valid at temperature 3.0. IR-gated model: 100% at all temperatures. Ablation: removing P_4 reduces validity by 16pp (P4-specific failures only); removing any other stage causes stage-specific degradation only.
- **OOD robustness.** Baseline: 0% valid on off-hours test (temporal OOD). Gated: 100%. Topology compiled from structure, not learned from data.
- **Valuation experiment.** Two edges: weight 5 ($v_5 = 1$) and weight 25 ($v_5 = 2$), both $\equiv 0 \pmod{5}$. FSM accuracy: 50%. Mod-5 gate: 50%. Valuation gate ($\tau = 2$): 100%. Non-Archimedean structure does real work.

What changes vs. prior work

Prior framing	This work
"Model is wrong"	"Which level of structure failed?"
Single-score hallucination rate	Three decomposed failure modes with distinct detectors
Post-hoc detection (after generation)	Pre-sampling support restriction (before softmax)
External oracle required	Admissibility compiled from structure; no oracle at inference
Probability error / retrieval mismatch	Projections of deeper structural failures in a semantic transition system

One sentence. Hallucinations can be decomposed into three mathematically distinct failure modes — global inconsistency, structural ambiguity, and generative instability — each detectable by different algebraic invariants ($H_1(\mathbb{Z}_5)$, $\mathbb{Z}_2$, 2-adic v_2) of the same semantic transition system, and preventable by pre-sampling support restriction before the softmax.

Full paper: `hallucination_paper.pdf` Code: `experiment4_z5_sanity.py`, `experiment5_minimal_geometry.py`, `benchmark_final.py`, `experiment3_padic.py`